FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: 1HA-I 24008 — 30083 (Version B) / 32081 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

MATHEMATICS HSSC-I — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 20 | **Paper Code:** 30083 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. What is the period of 3Sin(x/5)?

(A) 10π (B) 30π

(C) π/5 (D) 5π

Q2. Which of the given options state the solution of Sin x + Cos x = 0?

(A) -π/6 (B) -2π/3

(C) π/4 (D) 3π/4

Q3. What is the multiplicative inverse of -i?

(A) 1 (B) -1

(C) i (D) -i

Q4. The real part of (1+7i)/(3-4i) is:

(A) 1/2 (B) 1

(C) -1/2 (D) -1

Q5. If A is a matrix of order 3×2, then order of the product A^t A is:

(A) 3×2 (B) 3×3

(C) 2×2 (D) 2×3

Q6. If A = [i 0 0; 0 i 0; 0 0 i], then which one in the options is A^3?

(A) -iA (B) iA

(C) -A (D) A

Q7. What is the projection of a along b if a = 3i + j - k and b = i?

(A) 3/√1 (B) 3/√1 j

(C) 3/√11 (D) 3/√11 i

Q8. For what value of α, vectors 4i + 16j + αk and 2i + 8j - 4k are parallel to each other?

(A) 34 (B) 92

(C) -8 (D) 2

Q9. If x-3, 6, y+3 are in A.P then value of x+y is:

(A) 12 (B) 18

(C) 0 (D) 6

Q10. The nth term of a series 3×1^2 + 5×2^2 + 7×3^2 + ... is:

(A) (2n-1)n^2 (B) (2n+1)(n+1)^2

(C) (2n+1)n (D) (2n+1)n^2

Q11. In how many ways 5 persons can be seated at a round table?

(A) 4! (B) 1/2(4!)

(C) 5! (D) 1/2(5!)

Q12. What is the probability of drawing a King from a well shuffled pack of 52 playing cards?

(A) 13/52 (B) 4/52

(C) 1/52 (D) 2/52

Q13. What is the coefficient of 3rd term in the expansion of (x - 1/x)^8?

(A) ${}^8C_3$ (B) ${}^8C_4$

(C) 1 (D) ${}^8C_2$

Q14. Which one in the given options is true if 2^n > 2(n+1) ∀n ∈ Z^+?

(A) n < 3 (B) n > 3

(C) n < 2 (D) n > 2

Q15. The graph of y = x^4 is symmetrical about:

(A) x-axis (B) y-axis

(C) Origin (D) y = x

Q16. (-1,-1) is a solution of the inequality:

(A) 2x + y < -1 (B) 4x + 3y > 0

(C) -x - 2y < 0 (D) 2x - y > 1

Q17. Which of the following options equates Cos 196°?

(A) Sin 16° (B) -Sin 16°

(C) Cos 16° (D) -Cos 16°

Q18. If Cos β = 3/4, then value of Cos 2β is:

(A) 3√7/8 (B) -3√7/8

(C) 1/8 (D) -1/8

Q19. Area of a triangle ΔABC (with usual notations) a = 2, b = √3 and γ = π/3 is:

(A) 3/2 (B) 3

(C) 2 (D) √3/2

Q20. What is the shadow length of a √3 m high tree if sun's elevation is 45°?

(A) 1/√2 m (B) √3 m

(C) 1 m (D) 1/√3 m

MATHEMATICS HSSC-I — SECTIONS B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 80 | **Paper Code:** 1HA-I 24008 (B) (shared Sections B & C, versions B & D)

Note: SECTION – B (Marks 48) — Q.2 Solve the following Questions. (12 x 4 = 48)

درج سوالات حل کریں۔

(i) [04] Simplify z = (3+i)^3/(3-i) in the form a+ib where i = √-1 and find the value of |z|. — OR (04) — Find row rank of [1 2 3 2; 4 2 1 3; 5 2 -1 2]

(ii) [04] Solve the system of linear equations. (3-2i)x + (1+2i)y - 1 = 0, (3+2i)x - (1-2i)y - 1 = 0 — OR (04) — If 4th and 10th terms of a HP are 2/15 and 2/33 respectively, then find its 23rd term.

(iii) [04] If A = [5 9 2; 4 8 1; 3 7 0], then show that (A + A^t) is symmetric. — OR (04) — For what value of p, vectors 3pi + 1j - 5k and 2pi + pj + 2k are mutually perpendicular?

(iv) [04] Find the volume of a tetrahedron with vertices A(1,2,2), B(2,1,1), C(3,3,4) and D(0,1,5) — OR (04) — Insert four A.Ms between 5 and 25.

(v) [04] If 2nd and 6th terms of a GP are 3 and 3/4 respectively, find its 16th term. — OR (04) — Sum to n-terms the series 1.5 + 2.6 + 3.7 + 4.8 + ...

(vi) [04] How many 7-digit different numbers can be formed from the digits 5,5,6,6,9,9,9 using all and how many of them are greater than 9,950,000? — OR (04) — Prove that 1 + 4 + 7 + ... + (3n-2) = n(3n-1)/2 by using the mathematical induction.

(vii) [04] For a real valued function f(x) = (5x-2)/(x+2), x ≠ -2 find $f^{-1}(x)$ and determine its domain and range. — OR (04) — If Cos α = 3/5, Sin β = 5/13 with π/2 < β < π and 3π/2 < α < 2π, then find the value of Sin(α+β)

(viii) [04] State number of diagonals of an n-sided polygon and find number of diagonals of a nine sided polygon. — OR (04) — Prove that Sin 2θ + Sin 4θ + Sin 6θ + Sin 8θ = 4Sin 5θ Cos 2θ Cos θ

(ix) [04] Find the equation of a parabola y = ax^2 + bx + c that cuts x-axis at points (-4,0), (4,0) and passes through a point (0,8). — OR (04) — A pair of fair dice is thrown. The number of dots on the top are added. What is the probability of getting a sum greater than 9 or a sum divisible by 5.

(x) [04] Verify that Cos^4 θ = 1/8(3 + 2Cos 2θ + Cos 4θ) — OR (04) — Solve triangle ABC with α = 31°5', β = 50°55' and C = 13 cm using usual notations.

(xi) [04] Find radii of the escribed circles of triangle ABC opposite to the largest and smallest sides given that a = 13, b = 10 and c = 7 (using usual notations) — OR (04) — Without drawing, guess the graph of y = Sin θ/6 and find its period, frequency and amplitude.

(xii) [04] Verify that 2S = 8R Cos α/2 Sin β/2 Cos γ/2 — OR (04) — Verify that Tan^{-1} 3/4 - Tan^{-1} 4/3 + 2Tan^{-1} 1/7 = 0

> Note: SECTION – C (Marks 32) — Note: Solve the following Questions. (Use of graph paper is not necessary. Candidates can make their own grid on answer book) (4 x 8 = 32)

Q3. (Marks 08) Q.3 Find inverse of the matrix [1 1 2; 3 -1 1; -1 3 4] — OR (08) — If a = -10i + 2j + 4k and b = i - j + 2k then find a unit vector orthogonal to a × b. Also find angle between the vectors a and b.

Q4. (Marks 08) Q.4 Use Gauss Jordan method to solve the system of linear equations: x - 2y + z = 3; 3x + 5y = 11; 4y + 3z = 13 — OR (08) — If y = 1/(2(1!))(1/6) + 1.3/(4(2!))(1/6)^2 + 1.3.5/(8(3!))(1/6)^3 + ..., then verify that 5y^2 + 10y - 1 = 0

Q5. (Marks 08) Q.5 Find point of intersection of the functions f(x) = -x + 6 and g(x) = x^2 - 4x + 6 graphically. — OR (08) — Find general solution of a trigonometric equation 3Cos x + 3 = 2Sin^2 x

Q6. (Marks 08) Q.6 Find maximum and minimum values of a function f(x,y) = 2x + 3y subject to the constraints x + 2y ≤ 10, 3x + y ≤ 9, 9x + 8y ≤ 72, x ≥ 0, y ≥ 0 — OR (08) — Sketch the graph of y = 2Cos θ/2; -π ≤ θ ≤ π

VERSION D

MATHEMATICS HSSC-I — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 20 | **Paper Code:** 32081 (Version D)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. If z = -3 - 5i then value of z̄ - z is:

(A) 3 + 10i (B) 6

(C) 6 + 10i (D) 10i

Q2. What is the modulus of complex number (4+2i)/(1-2i)?

(A) 10/3 (B) 10/√5

(C) 2 (D) 4

Q3. For what value of x, [2 2 2; 0 x 6; 0 0 6] is a singular matrix?

(A) 0 (B) 1

(C) 1/12 (D) 12

Q4. If [1 2; 3 4] A = [1 0 2; 2 0 1; 3 1 2], then order of matrix A is:

(A) 2×2 (B) 2×3

(C) 3×2 (D) 3×3

Q5. For what value of α, vectors 3i + 4j - k and i - αj have the same magnitude?

(A) 25
(B) ±5
(C) √23
(D) 3/4

Q6. If |a.b| = |a×b| then angle between vectors a and b is:

(A) 30°
(B) 45°
(C) 90°
(D) 180°

Q7. For what value of q, numbers 1/3, 1/5, 1/q are in H.P?

(A) 1/7
(B) 1/15
(C) 7
(D) 15

Q8. n^{th} term of a series 2×1^2 + 4×2^2 + 6×3^2 + ... is:

(A) (2n)n^2
(B) (2n+1)n^2
(C) (2n+2)n^2
(D) 2n^2

Q9. In how many ways 5 keys can be arranged on a circular key ring?

(A) 5!
(B) 1/2(5!)
(C) 4!
(D) 1/2(4!)

Q10. What is the probability of getting one head on tossing two fair coins?

(A) 1/4
(B) 2/4
(C) 3/4
(D) 4/4

Q11. In the expansion of (x - 1/x)^11 6th term from end is:

(A) 5th
(B) 6th
(C) 7th
(D) 8th

Q12. For which given least value of n, the expression n! > (2^n - 1) remains true?

(A) 1
(B) 2
(C) 3
(D) 4

Q13. The graph of y = 2x^2 - 6x cuts x-axis at point(s):

(A) (0,0) only
(B) (3,0) only
(C) (0,0),(3,0)
(D) (0,0),(-3,0)

Q14. (0,0) is NOT a solution of the inequality:

(A) x + y + 1 > 0
(B) x - y < 1
(C) 2x + y < 1
(D) -2x + y + 1 < 0

Q15. Which one of the given options, equates Sin(254°)?

(A) Cos 16° (B) -Cos 16°

(C) Sin 16° (D) -Sin 16°

Q16. Which one of the given options, equates 2Sin 7x Sin 3x?

(A) Cos 10x - Cos 4x (B) Cos 10x + Cos 4x

(C) Cos 4x - Cos 10x (D) -Cos 4x - Cos 10x

Q17. What is the area of triangle ΔABC with a = 2, c = 4, β = π/6 (usual notations)

(A) 2 (B) 2√3

(C) 4 (D) 4√3

Q18. If shadow length of a building is 1/√3 times its height, the elevation angle of sun is:

(A) 30° (B) 45°

(C) 60° (D) 90°

Q19. What is the period of 5Sec(x/3)?

(A) 2π/3 (B) 3π

(C) 6π (D) 30π

Q20. One solution of the equation Cos x + Sin x = √2 is:

(A) -π/4 (B) π/4

(C) 2π/4 (D) 3π/4

MATHEMATICS HSSC-I — SECTIONS B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 80 | **Paper Code:** 1HA-I 24008 (D) (shared Sections B & C, versions B & D)

Note: SECTION – B (Marks 48) — Q.2 Solve the following Questions. (12 x 4 = 48)

درج سوالات حل کریں۔

(i) [04] Simplify z = (4-6i)(2+i)/((3+i)(1+i)) in the form a+ib where i = √-1 and find the value of |z|. — OR (04) — Find the value of x, if [x+1 2 3; 4 5 6; 7 8 x+9] is a singular matrix.

(ii) [04] Solve the system of linear equations. (2+3i)x + (1-i)y - 11 = 0, (1+i)x + (2-3i)y - 11 = 0 — OR (04) — A particle acted by constant force 2i + j, i + 2j, i + j + k and 3i - 2j + 4k is displaced from point P(0,-2,-3) to point Q(2,0,6). Find the work done.

(iii) [04] If A = [7 8 6; 6 7 5; 5 6 4], then show that (A - A^t) is skew-symmetric. — OR (04) — Find number of different arrangements that can be made from the letters of word PARALLELOGRAM (using all) and how many of these begin with PE and end with OM?

(iv) [04] Insert four G.Ms between 7 and 1701 — OR (04) — Find the volume of a parallelepiped with adjacent edges defined by vertices. A(0,1,2), B(1,2,1), C(5,5,6) and D(3,3,1)

(v) [04] The 11th and 19th terms of an AP are 42 and 74 respectively. Find the sum of first 20 terms of AP. — OR (04) — Sum to n-terms the series 1.2 + 2.3 + 3.4 + ...

(vi) [04] Verify that 3 + 7 + 11 + ... + (4n-1) = n(2n+1) by using the mathematical induction. — OR (04) — Find the values of 'n' and 'r' if ${}^{n}P_r = 15120$ and ${}^{n}C_r = 126$

(vii) [04] In an HP, 8th term is 2/5 and 17th term is 2/11. Find 35th term of the HP. — OR (04) — For a real valued function f(x) = (3x-2)/(x+4), find $f^{-1}(x)$ and determine its domain and range.

(viii) [04] In a single throw of two fair dice, the number of dots on the top are added. Find the probability of getting a sum of 7 or 9. — OR (04) — Without drawing, guess the graph of y = Cos(θ/6). Also find its period, frequency and amplitude.

(ix) [04] If Sec α = 13/4, Sec β = 13/5 with $3\pi/2 < \alpha < 2\pi$ and $3\pi/2 < \beta < 2\pi$, then find the value of tan(α+β). — OR (04) — Solve triangle ABC with a = 15, c = 20 and β = 60° using usual notations.

(x) [04] Verify that Cos 3θ + Cos 5θ + Cos 7θ + Cos 9θ = 4Cos θ Cos 2θ Cos 6θ — OR (04) — Find radii ('R' and 'r') of circumscribed and inscribed circles of triangle ABC with side measures a = 4cm, b = 7cm and c = 9cm (use usual notations).

(xi) [04] Verify that 2r = 8R Sin α/2 Sin β/2 Sin γ/2 (use usual notations). — OR (04) — Verify $\text{Sin}^4\theta = 1/8(3 + \text{Cos } 4\theta - 4\text{Cos } 2\theta)$

(xii) [04] Verify that $(\text{Sin}^{-1} 1/\sqrt{5} + \text{Sin}^{-1} 1/\sqrt{10}) + (\text{Cos}^{-1} 2/\sqrt{5} + \text{Cos}^{-1} 3/\sqrt{10}) = \pi/2$ — OR (04) — Find equation of a parabola y = ax^2 + bx + c (∀a,b,c∈R) that cuts x-axis at points (-5,0), (4,0) and passes through a point (1,18)

> Note: SECTION – C (Marks 32) — Note: Solve the following Questions. (Use of graph paper is not necessary. Candidates can make their own grid on answer book) (4 x 8 = 32)

Q3. (Marks 08) Q.3 Find inverse of the matrix [1 1 2; 3 -1 1; -1 3 4] — OR (08) — Find a vector of magnitude 14 units orthogonal to vectors a = -i + 3j and b = i + 2k both. Also find angle between the vectors a and b.

Q4. (Marks 08) Q.4 Use Gauss Jordan method to solve the system of linear equations. -x + y + 2z = 2; 3x - y + z = 6; -x + 3y + 4z = 4 — OR (08) — Find the point of intersection graphically from the following functions. f(x) = -x + 4; g(x) = x^2 - 3x + 1

Q5. (Marks 08) Q.5 If y = 1/((1!)2)(1/4) + 1.3/((2!)4)(1/4)^2 + 1.3.5/((3!)8)(1/4)^3 + ..., then prove that 3y^2 + 6y - 1 = 0 — OR (08) — Find general solution of a trigonometric equation Cos 2x = Sin x

Q6. (Marks 08) Q.6 Find maximum and minimum values of a function f(x,y) = 3x + 2y subject to the constraints $x + 2y \leq 8$, $5x - 2y \leq 10$, $7x - 5y \geq -35$, $x \geq 0$, $y \geq 0$ — OR (08) — Sketch the graph of $y = \text{Sin } \theta/2$; $-\pi \leq \theta \leq \pi$